

Event-Timing Priors: Temporal Point Process Likelihoods Correct the Long-Horizon Extreme-Event Statistics of Generative Surrogates

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July 2026

Abstract

Generative surrogates of chaotic dynamical systems — including modern diffusion-based weather and climate emulators — are routinely *evaluated* on extreme events but never *trained* on their timing. We show that this omission is consequential and correctable. Working with a two-scale Lorenz-96 testbed observed under realistic conditions (coarse emulator timestep, observation noise), we find that a conditional flow-matching surrogate reproduces marginal statistics well while systematically corrupting the *temporal* statistics of threshold-crossing extreme events in long rollouts: event rates inflate, inter-event-time distributions distort, and return periods drift. We introduce an *event-timing prior*: a neural temporal point process (TPP) fit to the event timings of a curated catalog, whose likelihood — evaluated on soft, differentiable events extracted from short surrogate rollouts — serves as an auxiliary fine-tuning signal. Naive likelihood maximization is mode-seeking and collapses the event rate; we instead use a likelihood-ratio objective against a co-trained “self-TPP” (the point-process analog of variational score distillation), whose gradient vanishes exactly when the surrogate’s event process matches the catalog’s. Across seeds, TPP-aux fine-tuning recovers inter-event-time distributions, burstiness, and return-period curves that invariant-measure matching [6] provably leaves unconstrained, while a rate-matched Poisson control shows the gains require the prior’s temporal *structure*, not merely its rate. A second system (Kuramoto–Sivashinsky) maps the method’s boundary: when rollout *dynamics* are deeply corrupted, a timing-only prior collapses the event rate rather than repairing it — a characteristic, easily monitored failure signature whose mechanism we analyze. Event-timing structure is thus a distinct, directly controllable axis of surrogate quality — one that complements, and does not substitute for, dynamics-level training signals.

1 Introduction

Data-driven surrogates now emulate weather and climate at a fraction of the cost of numerical models, and generative (diffusion-family) surrogates such as GenCast [18] explicitly target probabilistic skill. A striking asymmetry runs through this literature: extreme events — heat waves, cyclones, blocking episodes — dominate the *evaluation* of these models [8, 18, 24], yet no training objective in use says anything about *when* extremes occur. Standard objectives constrain pointwise forecast error or, at best, time-aggregated distributional statistics: invariant measures [6, 20], spectra [4], or adversarial OT statistics [15]. All of these are invariant to permuting time — they cannot distinguish a process whose extremes arrive quasi-periodically from one whose extremes arrive in bursts, so long as the marginals agree.

The temporal law of extremes is precisely what many downstream users care about: return periods drive design codes; clustering of exceedances drives compound risk; inter-event times drive recovery windows. And it is a quantity for which independent, high-quality data often exists even

when state observations are noisy: curated event catalogs (hurricane best tracks, earthquake catalogs, flood records) are compiled and quality-controlled separately from gridded reanalyses.

Contributions.

1. We demonstrate a clean dissociation: under realistic training conditions (coarse timestep, noisy observations), a strong generative surrogate’s long-horizon rollouts corrupt event-timing statistics in ways that invariant-measure/marginal-statistics training does not and cannot fix (§5).
2. We introduce TPP-aux fine-tuning: a neural TPP fit to catalog event timings acts as a differentiable prior on the timing of soft events extracted from short surrogate rollouts (§3). We identify and solve the mode-seeking pathology of naive TPP-likelihood maximization via a likelihood-ratio (self-TPP) objective, the point-process analog of variational score distillation [12, 23].
3. Controls that isolate the mechanism: a rate-matched *Poissonized* TPP prior (same loss form and strength, no temporal structure) actively damages event timing rather than repairing it; the invariant-measure objective of Jiang et al. [6] (given equally privileged clean statistics) excels at marginals but leaves return periods and burstiness unrepaired; pushforward-style stabilization [3] improves rate-side statistics without learning conditional timing structure, at a significant cost to spectra and probabilistic skill.
4. An honest map of the method’s boundary: on Kuramoto–Sivashinsky, where the base surrogate’s rollout *dynamics* are deeply corrupted (spectrum errors of e^3) despite near-perfect state marginals, every timing-only variant collapses the event rate — we analyze the mechanism (mode-seeking through the only accessible descent direction, outrunning the self-TPP brake) — and neither dynamics-level objectives nor naive composition fully recovers the event process (§5.2).

2 Related work

Long-time statistics of neural surrogates. Jiang et al. [6] train neural operators to preserve invariant measures of chaotic attractors via an optimal-transport (Sinkhorn) loss on physics-informed summary statistics, or a contrastive feature loss; DySLIM [20] regularizes toward the invariant measure; spectral losses match frequency content [4]; adversarial OT regularization matches trajectory statistics [15]. These objectives are functions of the *time-marginal* (or time-averaged) law of the process and are therefore blind to the ordering and spacing of rare events; we use Jiang et al. [6] as our primary control. Rollout stabilization — the pushforward trick [3], noise injection and refinement [10, 17], Jacobian regularization [16] — targets stability, an orthogonal axis.

Extremes and generative emulators. Stamatelopoulos and Sapsis [22] study when diffusion models extrapolate amplitude tails; Manshausen et al. [13] extract extreme-event likelihoods from a trained diffusion emulator at inference time. Flagship emulators evaluate extremes diagnostically [8, 18, 24]. None of these place event *timing* in the training signal. Raeth and Ludwig [19] augment generative forecasters with differentiable decision losses — structurally the closest training-objective precedent — but target decision cost, not temporal event structure, and no dynamical rollout is involved.

Neural temporal point processes. Neural TPPs model conditional intensities of event sequences [14] or inter-event densities directly [21]; see [1] for benchmarking. To our knowledge, TPP likelihoods have not previously been used as training objectives for dynamical-system surrogates.

3 Method

3.1 Setup: surrogates, events, catalogs

Let $\mathbf{x}_t \in \mathbb{R}^K$ be the observed (slow) state of a chaotic system sampled at interval Δt . A generative surrogate models $p_\theta(\mathbf{x}_{t+1} | \mathbf{x}_t)$; long rollouts compose samples autoregressively. We train on noisy observations $\tilde{\mathbf{x}}_t = \mathbf{x}_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, r^2 \sigma_x^2 I)$, reflecting the practical setting where state data comes from noisy analyses [6].

Events are threshold upcrossings of a scalar site observable $s_k(t)$ (here $s_k = -x_k$, deep troughs; threshold $u =$ the pooled 95th percentile): an event occurs at step t of stream k iff $s_k(t) > u$ and $s_k(t-1) \leq u$. By rotational symmetry the K site streams are exchangeable. We assume access to a *catalog*: the event timings extracted from the clean historical record (in practice, curated event databases; in our testbed, events of the clean trajectories underlying the noisy training data). Crucially, the catalog contains *event times only* — orders of magnitude less information than the full state record.

3.2 The event-timing prior: a recurrent-intensity TPP

We fit a discrete-time recurrent-intensity TPP p_{ϕ^*} to catalog streams: a GRU consumes the binary event sequence $w_{1:T}$ of one stream and emits a per-step conditional intensity $\lambda_t = \text{softplus}(g(h_{t-1}))$ (events per unit time, depending only on events strictly before t), with log-likelihood

$$\mathcal{L}(w; \lambda) = \sum_t [w_t \log \lambda_t - \lambda_t \Delta t], \quad (1)$$

the standard grid discretization of the continuous point-process likelihood [5, 14], exact as $\Delta t \rightarrow 0$. Two properties matter. First, (1) is *linear in w_t* , so relaxed (soft) event indicators substitute cleanly — this is what makes the prior differentiable end-to-end. Second, the same fitted model evaluates any candidate event stream, which is how it becomes a training signal.

3.3 Soft events from differentiable rollouts

From a rollout $\hat{\mathbf{x}}_{0:T}$ we extract soft upcrossing weights

$$w_t = \sigma((s_t - u)/\tau) \sigma((u - s_{t-1})/\tau) \in [0, 1], \quad (2)$$

with temperature τ calibrated so the soft event rate matches the hard rate on held-out data (bias $< 5\%$ at our τ). The surrogate’s few-step deterministic sampler makes each rollout step differentiable given its noise draw, and gradients through the chaotic dynamics are truncated to windows of T_w steps (states are recorded in a gradient-free rollout, and all windows are re-forwarded with gradients in parallel from their recorded start states — identical values and gradient semantics to truncated BPTT, at a fraction of the sequential cost). The TPP consumes the full stream: each rollout is prefixed with the catalog’s event history preceding the initial condition, so intensities are history-calibrated where the likelihood begins to count.

3.4 Mode-seeking, and the likelihood-ratio fix

The obvious auxiliary loss, $-\mathbb{E}[\mathcal{L}(w^{\text{soft}}; \lambda_{\phi^*})]$, is a density, not a fit measure: under a fitted TPP the *empty* sequence often has higher density than typical sequences, so maximizing catalog-TPP likelihood of rollout events drives the surrogate toward the TPP’s mode — suppressing events rather than matching their law. We observe exactly this collapse empirically (§6). The fix mirrors

variational score distillation [12, 23]: maintain a small *self-TPP* p_ψ , refit continually by maximum likelihood on the surrogate’s own (hard, detached) rollout events, and optimize the ratio

$$\mathcal{L}_{\text{aux}}(\theta) = -\mathbb{E}_{\text{rollouts}}[\mathcal{L}(w^{\text{soft}}; \lambda_{\phi^*}) - \mathcal{L}(w^{\text{soft}}; \lambda_\psi)]. \quad (3)$$

In expectation this is the reverse KL between the surrogate’s event process and the catalog process (up to the self-TPP’s fit error): its gradient vanishes exactly when the two laws agree, and the self-TPP term cancels the uniform “fewer events are more likely” pressure. Fine-tuning minimizes $\mathcal{L}_{\text{FM}} + \lambda_{\text{aux}}\mathcal{L}_{\text{aux}}$, keeping the base flow-matching loss as an anchor on ordinary skill.

3.5 Conditions

All conditions fine-tune the *same* per-seed base checkpoint for the same number of steps; they differ only in the auxiliary objective: **base** (flow-matching only, compute-matched); **TPP-aux** (ours, Eq. 3 with the catalog TPP); **Poisson-aux** (identical, but the prior TPP is fit to Poissonized catalogs — same rate, exponential inter-event times; the content-specificity control); **invariant-stats** (the OT objective of Jiang et al. [6]: debiased Sinkhorn divergence between rollout and clean-climatology clouds of their physics-informed summary statistics $\{\dot{u}_i, (u_{i+1} - u_{i-2})u_{i-1}, u_i\}$ — given the same clean-reference privilege as the catalog); **pushforward** [3]; **deterministic** (MSE-trained autoregressive baseline).

4 Experimental setup

System. Two-scale Lorenz-96 [11, 25]: $K=40$ slow sites, $J=10$ fast variables per site ($F=10$, $h=1$, $b=c=10$), RK4 at $dt=0.002$, observed at $\Delta t = 0.2$ MTU (coarse emulator step). Training: 64 trajectories $\times$ 160 MTU with observation noise $r=0.3$; evaluation references from clean held-out runs (26,600 MTU total).

Ground-truth event structure. Trough events are strongly sub-Poisson (quasi-periodic recurrence): IET CV 0.78, Fano factors 0.6–0.7 across windows; near-renewal (lag-1 IET correlation ≈ 0). Aggregate (spatial-max) events are super-Poisson (CV 1.15) — used as a held-out transfer diagnostic.

Models. Surrogate: conditional flow matching [9] on normalized state residuals; 1D circular CNN (4 residual blocks, width 96, FiLM noise-level conditioning, $\sim 0.66\text{M}$ parameters); 6-step Euler sampler. TPP: single-layer GRU, hidden 64 ($\sim 40\text{k}$ parameters). Full hyperparameters in appendix.

Metrics. Event timing: pooled IET distribution (two-sample KS, W_1), event rate ratio, Fano-factor curve error, return-period curves at catalog quantiles $q \in \{0.9, \dots, 0.995\}$, catalog-TPP held-out log-likelihood of rollout events. Ordinary skill: pooled state W_1 , PSD log-distance, ACF error, short-horizon CRPS and ensemble spread. Stability: diverged-rollout fraction (censored and reported). ≥ 5 confirmatory seeds, paired by shared base checkpoint; the tuning seed is excluded.

5 Results

5.1 Lorenz-96: the event-timing deficit and its repair

The base surrogate corrupts event timing. Trained on noisy observations, the base surrogate’s long rollouts (64×800 MTU) inflate the event rate by $10.0\% \pm 1.2\%$, distort the inter-event-time distribution (IET KS 0.065 ± 0.007 , W_1 0.41 ± 0.04 MTU), and shift return periods (mean $|\log \text{RP}_{\text{model}}/\text{RP}_{\text{true}}| = 0.105 \pm 0.009$ across catalog quantiles 0.90–0.995). An identically-budgeted

Table 1: Lorenz-96 main results: mean $\pm$ s.d. over confirmatory seeds (shared base checkpoint per seed; identical fine-tuning budgets). Bold = best among fine-tuned generative conditions. IET = inter-event time; RP = return period; TPP-LL = catalog-TPP log-likelihood of rollout events (ground-truth ceiling -0.0994); CRPS at lead 20 steps.

condition	IET KS	IET W_1	rate ratio	RP log-err	TPP-LL	state W_1	CRPS@20
Base (FM only)	0.065 ± 0.007	0.414 ± 0.044	1.100 ± 0.012	0.105 ± 0.009	-0.1018 ± 0.0001	0.094 ± 0.014	1.735 ± 0.002
TPP-aux (ours)	0.028 ± 0.004	0.142 ± 0.068	1.005 ± 0.035	0.055 ± 0.033	-0.0992 ± 0.0012	0.066 ± 0.045	1.743 ± 0.008
Poisson-TPP-aux	0.095 ± 0.018	1.054 ± 0.215	0.809 ± 0.032	0.229 ± 0.075	-0.0946 ± 0.0013	0.153 ± 0.032	1.769 ± 0.014
Invariant-stats	0.030 ± 0.005	0.211 ± 0.087	0.965 ± 0.034	0.117 ± 0.043	-0.0971 ± 0.0015	0.048 ± 0.019	1.755 ± 0.019
Pushforward	0.047 ± 0.003	0.142 ± 0.023	1.029 ± 0.007	0.036 ± 0.006	-0.1011 ± 0.0002	0.076 ± 0.005	1.767 ± 0.003
Deterministic AR	0.173 ± 0.082	1.021 ± 0.453	1.310 ± 0.197	0.454 ± 0.246	-0.1030 ± 0.0117	0.473 ± 0.151	3.225 ± 0.023

continuation of flow-matching training (**base** condition) leaves all of these unchanged — the deficit is persistent, not undertraining.

TPP-aux repairs it without costing skill. Fine-tuning with the event-timing prior recovers the event rate (1.005 ± 0.035 ; paired t -test vs. base $p = 0.006$), the IET distribution (KS 0.028 ± 0.004 , $p = 2 \times 10^{-4}$; W_1 0.14 ± 0.07 , $p = 0.001$), and return periods (0.055 ± 0.033 , $p = 0.038$). The catalog-TPP log-likelihood of rollout events rises to -0.0992 ± 0.0012 — statistically indistinguishable from the value the catalog TPP assigns to *real* held-out event sequences (-0.0994): the fine-tuned surrogate’s events are exactly as typical under the prior as truth itself, neither less (deficit) nor more (mode-seeking). Short-horizon CRPS changes by $+0.4\%$ ($p = 0.16$); no rollout diverged in any condition (Table 1).

Structure, not rate: the Poisson control. The Poissonized prior — identical loss form, identical strength, identical rate information, exponential inter-event times — does not merely fail; it actively damages the surrogate (IET W_1 1.05 ± 0.21 , $p = 0.004$; rate ratio 0.81 ± 0.03 , $p = 3 \times 10^{-5}$; Fano error $+0.058$, $p = 7 \times 10^{-5}$). Its rollouts’ hazard function is driven toward a constant (Fig. 2), erasing the quasi-periodic recurrence structure the dynamics should carry: optimizing toward a structureless event law distorts the dynamics that produce events. The gains of TPP-aux are therefore attributable to the *temporal structure* of the prior, not to rate matching or to the loss mechanics.

What the timing prior transfers. Ground-truth trough events are quasi-periodic: the IET distribution is multimodal at harmonics of the wave-recurrence period, and the conditional hazard oscillates (Fig. 2, black). The base surrogate over-amplifies the hazard oscillation; TPP-aux pulls it onto the ground-truth curve; the Poisson prior flattens it. On the IET quantile–quantile plot (Fig. 1), TPP-aux is the only condition tracking the diagonal across the full range.

Comparison with time-marginal objectives. The invariant-statistics control [6], given clean summary-statistic clouds of equal privilege, achieves the best state marginals and spectra (as designed) and — in this noise-dominated regime, where the base’s timing errors are largely inherited from amplitude corruption — also improves IET-level statistics. But it does not improve return periods (0.117 ± 0.043 vs. base 0.105 , $p = 0.61$), significantly worsens burstiness (Fano error $p = 0.023$), undershoots the event rate (0.965), and its rollout events are *more* typical than truth under the catalog TPP (-0.0971 , overshooting the ceiling). The pushforward control improves rate-side statistics (return periods 0.036 ± 0.006) — consistent with its denoising effect on rollouts — but leaves the conditional timing structure untouched (catalog-TPP LL -0.1011 , at base level) and significantly degrades spectra ($p = 10^{-5}$) and CRPS ($p = 5 \times 10^{-5}$). Only TPP-aux exhibits the full pattern: conditional timing at the ground-truth ceiling, marginals improved, and ordinary skill preserved. The deterministic AR baseline, for reference, is far worse on every axis (CRPS 3.23 vs. 1.74 ; state W_1 0.47), confirming that generative surrogates are the right object of study.

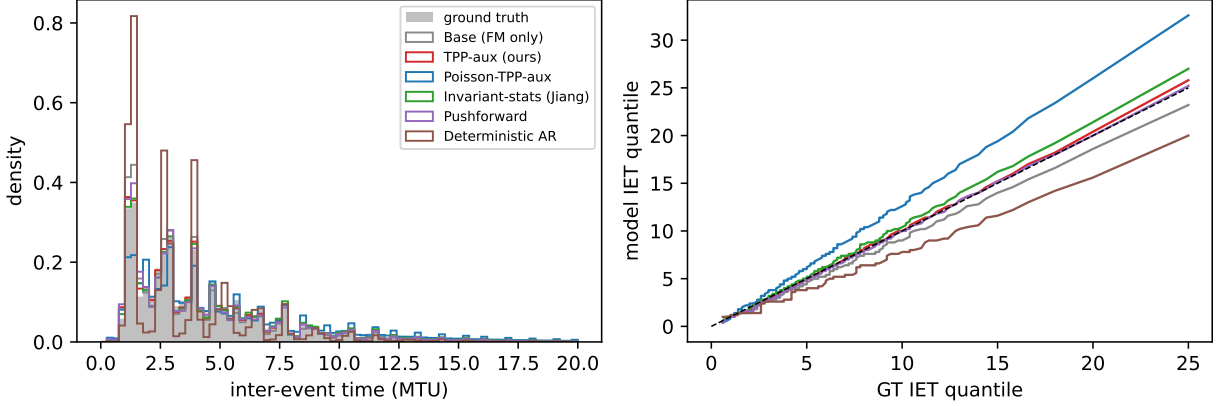


Figure 1: Inter-event-time statistics, Lorenz-96 (pooled over 5 seeds). Left: IET densities; ground truth (shaded) is multimodal at wave-recurrence harmonics. Right: quantile–quantile against ground truth; TPP-aux (red) tracks the diagonal; the Poissonized prior (blue) severely distorts the distribution.

5.2 Kuramoto–Sivashinsky: the boundary of a timing-only prior

KS ($L = 22$, $N = 64$, $\Delta t = 1$ tu; events = upcrossings of u at its 95th percentile, *clustered*, IET CV 1.58 — the opposite deviation from Poisson to Lorenz-96) probes a qualitatively different deficit. The noisy-observation base surrogate’s rollouts have a nearly perfect state marginal (state $W_1 = 0.037$) but a severely corrupted event process ($2.5\times$ the true rate; IET KS 0.59; return-period log-error 1.14) driven by a *deep dynamical* corruption: the rollout power spectrum is wrong by e^3 at high frequencies — jittery dynamics generating spurious threshold re-crossings. This is the setting the introduction’s logic points to: nothing here is fixable by matching state marginals alone.

The outcome inverts Lorenz-96, and is instructive. **The timing-only prior fails, by a characteristic mechanism.** Under TPP-aux the event rate collapses to 0.31 of truth (IET W_1 worsens from 55 to 180), and the collapse persists across auxiliary weights $\{1, 3, 10\}$, self-TPP budgets $\{5, 25\}$ steps, gentler and $3\times$ longer fine-tuning, and a milder noise level ($r=0.2$; rate 0.19). The training traces show why: a clustered prior assigns very low baseline intensity between bursts, so each of the base model’s many mistimed events is heavily penalized, and within ~ 200 steps the surrogate’s event process shoots *past* the target into sparsity while the self-TPP lags; once the self-TPP catches up, the two likelihoods equilibrate ($\mathcal{L}_{\phi^*} \approx \mathcal{L}_{\psi}$, both above the ground-truth ceiling) and the residual KL pressure ($\approx 0.005/\text{step}$) is too weak against the flow-matching anchor to escape. In short: when the underlying dynamics are too corrupted to offer a gradient direction that *places* events correctly, suppressing events is the accessible descent path, and the ratio objective’s correct fixed point is unreachable. Suppression—not repair—is the failure signature to monitor (the rollout event rate makes it visible immediately).

The controls invert accordingly (Table ??). The rate-matched *Poisson* prior — harmful on Lorenz-96 — is the best return-period condition on KS (RP log-error 0.16; rate 0.96): where the dominant error *is* the rate, a rate-only prior transfers exactly the information that is broken, and its permissive constant hazard exerts no placement pressure that the corrupted dynamics cannot satisfy. The two systems are mirror images: a prior helps precisely to the extent that its content matches the deficit, and a structured prior whose structure the dynamics cannot realize does damage instead.

What does repair KS’s temporal structure is a dynamics-level statistic. The Jiang et al. [6] objective — whose KS summary statistics include the *temporal increment* u_t , a local-dynamics

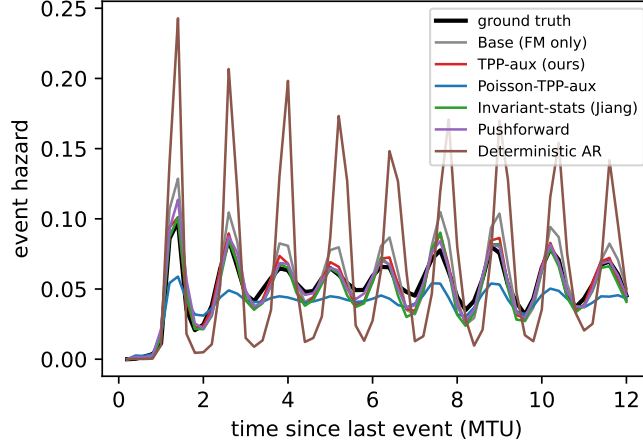


Figure 2: Conditional event hazard vs. time since the last event. Ground truth (black) oscillates at the recurrence period. The base surrogate over-amplifies the oscillation; TPP-aux (red) matches it; the rate-matched Poisson prior (blue) flattens it — the content-specificity dissociation in a single curve.

Table 2: Kuramoto–Sivashinsky results (3 seeds; †: single exploratory seed). Ground-truth catalog-TPP ceiling: -0.050 . The timing-only prior collapses the event rate; the rate-only (Poisson) prior — harmful on Lorenz-96 — is here the best return-period condition; increment statistics (u_t) repair the dynamics and most of the timing.

condition	IET KS	IET W_1	rate ratio	RP log-err	TPP-LL	state W_1	PSD log-dist
Base (FM only)	0.594 ± 0.002	54.6 ± 0.0	2.50 ± 0.00	1.14 ± 0.00	-0.162 ± 0.000	0.037 ± 0.001	3.40 ± 0.00
TPP-aux	0.464 ± 0.128	179.5 ± 11.6	0.31 ± 0.01	1.92 ± 0.18	-0.025 ± 0.001	0.234 ± 0.027	3.15 ± 0.01
Poisson-TPP-aux	0.232 ± 0.011	35.7 ± 1.0	0.96 ± 0.03	0.16 ± 0.01	-0.065 ± 0.002	0.156 ± 0.034	3.04 ± 0.05
Invariant-stats (incl. u_t)	0.262 ± 0.025	22.3 ± 5.1	1.03 ± 0.10	0.66 ± 0.13	-0.069 ± 0.006	0.075 ± 0.017	1.83 ± 0.02
Invariant-stats + TPP-aux †	0.271	44.5	0.70	1.28	-0.049	0.175	1.92

constraint rather than a pure time-marginal — repairs the spectrum ($3.40 \rightarrow 1.83$) and with it most of the event statistics (rate 1.03, IET KS 0.26), though substantial timing error remains (IET W_1 22 vs. a base of 55; Fano error 0.34): no condition fully recovers the KS event process. Composing the two priors (climatology + catalog) reaches the correct typicality level (catalog-TPP LL at the ceiling) and halves the base IET KS, but the timing term still drags the rate down (0.70) relative to the increment objective alone — the priors do not simply add.

Together the two systems bracket the method: when a surrogate’s dynamics are structurally sound and its event timing is corrupted (Lorenz-96), a catalog TPP prior repairs the event process completely and specifically; when the dynamics themselves are deeply corrupted (KS), event-timing information alone cannot substitute for dynamics repair — it must wait on, or be combined with, objectives that restore the dynamics, and naive combination is not enough. We regard characterizing — and lifting — this boundary as the central open problem this paper leaves.

6 Ablations

Naive likelihood collapses. Replacing the ratio objective with plain likelihood maximization ($-\mathbb{E}[\log p_{\phi^*}]$, no self-TPP) suppresses 99.75% of events (rate ratio 0.0025) while *increasing* the

catalog-TPP log-likelihood of rollouts to -0.055 , far above the ground-truth ceiling of -0.0994 : near-empty sequences really are the density mode of a fitted TPP, and a surrogate optimized toward that mode stops producing extremes at all. The self-TPP ratio term is not a refinement but a requirement.

Which structure matters: the shuffled control. A prior fit to interval-*shuffled* catalogs (IET marginal preserved, serial correlations destroyed) performs on par with the full prior (IET KS 0.022, rate 0.995, seed 0), consistent with the near-renewal character of the ground-truth event process (lag-1 IET correlation ≈ 0). Together with the Poisson control this brackets the operative content of the prior: the *shape of the inter-event-time distribution* — destroyed by Poissonization, preserved by shuffling — is what repairs the surrogate.

Soft-event temperature. At our τ the soft event rate is biased $+4.9\%$ with 80% of soft mass on true hard-event steps; doubling τ inflates the bias to 17.5% (appendix).

No free transfer across event definitions (negative result). Site-level TPP-aux does not repair the statistics of a *different* event process — upcrossings of the spatial maximum (clustered, CV 1.13): the aggregate event rate is not restored (ratio 1.45 ± 0.22 vs. base 1.29 ± 0.04) and no condition recovers the aggregate clustering. The prior repairs the event process it was fit to; observables of interest each need their own catalog term. This sharpens the interpretation: TPP-aux is a targeted prior on a specified event law, not a generic rollout regularizer.

7 Discussion and limitations

Why fine-tuning with a tiny prior works. The catalog TPP has $\sim 40k$ parameters and sees only event times — a few thousand numbers — yet repairs a surrogate trained on six orders of magnitude more state data. Event timing is a low-dimensional, data-efficient summary of exactly the aspect of the dynamics that ordinary training leaves unconstrained under observation noise. This information asymmetry mirrors practice: curated event catalogs (best tracks, earthquake catalogs, flood records) are compiled independently of — and are far cleaner than — gridded state estimates.

Limitations. (i) Our systems are canonical chaotic testbeds, not full emulator stacks; scale remains to be demonstrated. (ii) The catalog is assumed accurate; catalog noise/censoring would enter the prior. (iii) The TPP likelihood is grid-discretized at the emulator step (exact only as $\Delta t \rightarrow 0$). (iv) Truncated rollout gradients bias the aux gradient; DDPO-style score-function estimators [2] would remove that bias at higher variance — we found the biased low-variance path sufficient where the method works, and no estimator choice rescues the KS regime, whose failure is a property of the optimization landscape, not the gradient estimator. (v) The method’s domain of applicability — dynamics sound, timing corrupted — must be diagnosed per application; the rollout event rate during fine-tuning is a cheap, reliable canary for leaving it. (vi) In noise-dominated regimes, part of the timing deficit is marginal-reducible and strong time-marginal objectives recover that part; our controls quantify exactly the residual that they cannot.

References

- [1] Yahya Aalaila, Gerrit Großmann, and Sebastian Vollmer. Seahorse: A unified benchmarking framework for spatiotemporal event modeling. *arXiv preprint arXiv:2607.01022*, 2026.
- [2] Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning. In *International Conference on Learning Representations*, 2024.

- [3] Johannes Brandstetter, Daniel Worrall, and Max Welling. Message passing neural PDE solvers. In *International Conference on Learning Representations*, 2022.
- [4] Dibyajyoti Chakraborty, Arvind T. Mohan, and Romit Maulik. Binned spectral power loss for improved prediction of chaotic systems. *Journal of Computational Physics*, 558:114866, 2026. doi: 10.1016/j.jcp.2026.114866.
- [5] Daryl J. Daley and David Vere-Jones. *An Introduction to the Theory of Point Processes: Volume I: Elementary Theory and Methods*. Springer, 2003.
- [6] Ruoxi Jiang, Peter Y. Lu, Elena Orlova, and Rebecca Willett. Training neural operators to preserve invariant measures of chaotic attractors. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [7] Aly-Khan Kassam and Lloyd N. Trefethen. Fourth-order time-stepping for stiff PDEs. *SIAM Journal on Scientific Computing*, 26(4):1214–1233, 2005.
- [8] Dmitrii Kochkov, Janni Yuval, Ian Langmore, Peter Norgaard, Jamie Smith, Griffin Mooers, Milan Klöwer, James Lottes, Stephan Rasp, Peter Düben, Sam Hatfield, Peter Battaglia, Alvaro Sanchez-Gonzalez, Matthew Willson, Michael P. Brenner, and Stephan Hoyer. Neural general circulation models for weather and climate. *Nature*, 632:1060–1066, 2024. doi: 10.1038/s41586-024-07744-y.
- [9] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- [10] Qi Liu, Laure Zanna, and Joan Bruna. Towards infinitely long neural simulations: Self-refining neural surrogate models for dynamical systems. *arXiv preprint arXiv:2603.17750*, 2026.
- [11] Edward N. Lorenz. Predictability: A problem partly solved. In *Proc. Seminar on Predictability*, volume 1. ECMWF, 1996.
- [12] Weijian Luo, Tianyang Hu, Shifeng Zhang, Jiacheng Sun, Zhenguo Li, and Zhihua Zhang. Diff-Instruct: A universal approach for transferring knowledge from pre-trained diffusion models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [13] Peter Manshausen, Noah Brenowitz, Jan Berner, Karthik Kashinath, and Michael Pritchard. Towards accurate extreme event likelihoods from diffusion model climate emulators. *arXiv preprint arXiv:2605.03802*, 2026.
- [14] Hongyuan Mei and Jason Eisner. The neural Hawkes process: A neurally self-modulating multivariate point process. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [15] André Melo, Luis Santiago, and Peter Lu. Learning to emulate chaos: Adversarial optimal transport regularization. *arXiv preprint arXiv:2604.21097*, 2026.
- [16] Fengxiang Nie and Yasuhiro Suzuki. Jacobian-adaptive weighting for stability: Enhancing long-term rollout of neural partial differential equation solvers via spatially-adaptive regularization. *arXiv preprint arXiv:2603.05538*, 2026.

- [17] Chris Pedersen, Laure Zanna, and Joan Bruna. Thermalizer: Stable autoregressive neural emulation of spatiotemporal chaos. In *International Conference on Machine Learning*, 2025. arXiv:2503.18731.
- [18] Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R. Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, Remi Lam, and Matthew Willson. Probabilistic weather forecasting with machine learning. *Nature*, 637:84–90, 2025. doi: 10.1038/s41586-024-08252-9.
- [19] Kornelius Raeth and Nicole Ludwig. Decision-aware training for sample-based generative models. *arXiv preprint arXiv:2607.01171*, 2026.
- [20] Yair Schiff, Zhong Yi Wan, Jeffrey B. Parker, Stephan Hoyer, Volodymyr Kuleshov, Fei Sha, and Leonardo Zepeda-Núñez. DySLIM: Dynamics stable learning by invariant measure for chaotic systems. *arXiv preprint arXiv:2402.04467*, 2024. ICML 2024.
- [21] Oleksandr Shchur, Marin Biloš, and Stephan Günnemann. Intensity-free learning of temporal point processes. In *International Conference on Learning Representations*, 2020.
- [22] Stamatis Stamatelopoulos and Themistoklis P. Sapsis. Can diffusion models capture extreme event statistics? *Computer Methods in Applied Mechanics and Engineering*, 435:117589, 2025. doi: 10.1016/j.cma.2024.117589.
- [23] Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan Li, Hang Su, and Jun Zhu. ProlificDreamer: High-fidelity and diverse text-to-3D generation with variational score distillation. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [24] Oliver Watt-Meyer, Brian Henn, Jeremy McGibbon, Spencer K. Clark, Anna Kwa, W. Andre Perkins, Elynn Wu, Lucas Harris, and Christopher S. Bretherton. ACE2: accurately learning subseasonal to decadal atmospheric variability and forced responses. *npj Climate and Atmospheric Science*, 8:205, 2025. doi: 10.1038/s41612-025-01090-0.
- [25] Daniel S. Wilks. Effects of stochastic parametrizations in the Lorenz ’96 system. *Quarterly Journal of the Royal Meteorological Society*, 131(606):389–407, 2005.

A Experimental details

Data. Two-scale Lorenz-96: RK4 at $dt=0.002$ (validated against 0.001: climatological mean/std/ q_{98} agree to $< 0.3\%$), observed every $\Delta t=0.2$ MTU after 20 MTU burn-in. Train: 64 trajectories $\times$ 160 MTU (51,136 transitions); val: 16 $\times$ 160; evaluation references: 64 $\times$ 400 MTU plus 8 $\times$ 1500 MTU held-out clean runs. Observation noise $\mathcal{N}(0, (0.3\sigma_x)^2)$ applied to train and val only. KS: ETDRK4 [7] at $dt=0.25$, $L=22$, $N=64$, observed every 1 tu; splits proportional.

Event definitions. L96: upcrossings of $s_k = -x_k$ above the pooled clean 95th percentile ($u = 3.082$); soft temperature $\tau = 0.025 \sigma_x = 0.088$ (soft/hard rate bias +4.9%, 80% of soft mass on hard-event steps; 2τ : +17.5%/62%; 4τ : +53%/38%). KS: upcrossings of u_i above its 95th percentile. Ground-truth structure at the emulator step: L96 troughs sub-Poisson (IET CV 0.78; lag-1 IET correlation ≈ 0 ; $\text{Fano}_2/\text{Fano}_{20} = 0.71/0.62$); KS crests super-Poisson (CV 1.58, $\text{Fano}_{50} \approx 1.11$).

Models and training. Surrogate backbone: stem + 4 circular-conv residual blocks (width 96, kernel 5, GroupNorm, FiLM conditioning on the flow time), zero-init head; $\approx 0.66\text{M}$ parameters;

residual parameterization $g = (x_{t+1}^n - x_t^n)/\sigma_d$. Base training: 20k steps, AdamW, lr 3×10^{-4} , batch 512, EMA 0.999. Fine-tuning (all conditions): 800 steps, lr 10^{-4} , EMA 0.995, $\lambda_{\text{aux}} = 10$ (TPP-family), $\lambda_{\text{marg}} = 1$; selected on a development seed excluded from all reported statistics; the Poisson control uses the tuned TPP settings unchanged. Sampler: 6 Euler steps. TPP: 1-layer GRU (hidden 64) + 2-layer head, softplus intensity; fit 3k steps on catalog streams (windows 512, 64-step warmup masked).

Auxiliary rollouts. 16 rollouts $\times$ 100 steps per aux application (every 2nd optimizer step), initialized at training states, prefixed with 100 steps of catalog event history (likelihood masked over the prefix). Gradients truncated to 25-step windows: states and noises are recorded in a gradient-free rollout, then all windows re-forward in parallel with gradients from their recorded starts — identical values and gradient semantics to truncated BPTT at a fraction of the sequential cost. The gradient-free rollout is executed as a captured CUDA graph. Self-TPP: initialized from the Poissonized TPP, warm-started on 40 init-model rollout batches, then 5 Adam steps (lr 3×10^{-3}) per aux application on the surrogate’s hard rollout events. Rollouts with $\max_k |x^n| > 8$ are censored from the aux loss (none occurred in reported runs).

Evaluation. 64 rollouts $\times$ 4000 steps from held-out clean initial conditions per checkpoint; censoring at 8σ (no rollout was censored in any reported condition). Return periods computed as pooled stream-time per upcrossing at clean thresholds $q \in \{0.90, 0.95, 0.98, 0.99, 0.995\}$. CRPS: 256 initial conditions, 16-member ensembles. Hazard curves: pooled empirical discrete hazard, bins with ≥ 50 at-risk intervals.

Compute. All experiments on one RTX 3080 (10 GB). Base training ≈ 9 min; TPP-family fine-tune ≈ 20 min; the full L96 confirmatory grid ≈ 8 h.